Strings - DeepDrive / Part-2

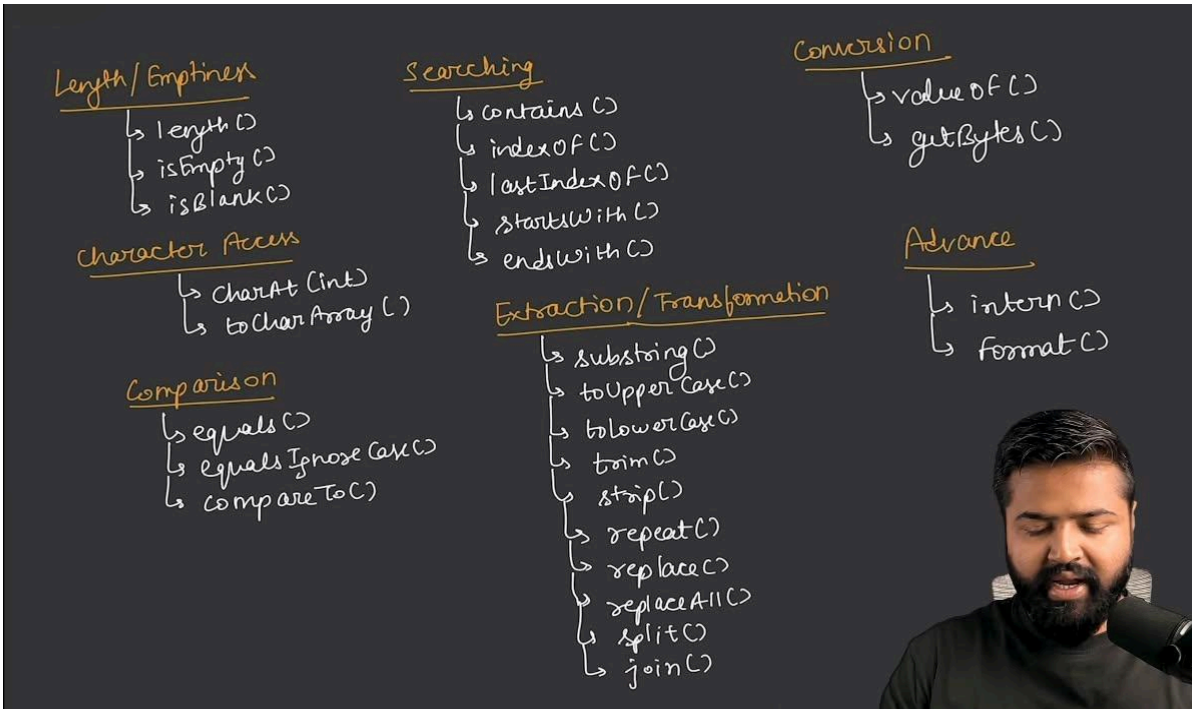
Internally String class→

Public final class String {

Instance variables→

private final byte[ ] values;
private final byte coder;
private int hash; }

Important methods in string class→




```java
String s1 = new String ("Priyansh");
String s2 = new String("Priyansh");

// Length /Emptiness.
    System.out.println(s1.length());//8
    System.out.println(s1.isEmpty());// false--> agar sirf "spaces" hoge to bhi ye false hi hoga.
    System.out.println(s1.isBlank());//false---> agar sirf "spaces" hoge to ye true hoga.
// Character accrss.
    System.out.println(s1.charAt(2));//"i"--> index value of the 2nd index.
    System.out.println(s1.toCharArray());// Priyansh--> char[] arr = s.toCharArray();
// Comparison Methods.
    System.out.println(s1.equals(s2));//true--> case sensitive.
    System.out.println(s1.equalsIgnoreCase(s2)); //true--> it ignore the cases of letters.'A','a'
    System.out.println(s1.compareTo(s2));/* it performs lexicographical comparison-->
    a dictionary comparison, matlab--> "saurav" / "sairav" --> "sairav" phle aayega.
    -> this method=>gives -ve[s1<s2], 0[both equal], +ve[s1>s2] outputes dega.
    ->s1 or s2 ki ASCII values ko add krke totals ko subtract kr dege.              */
// Searching.
    System.out.println(s1.contains("ansh"));// true--> it is case sensative.
    System.out.println(s1.indexOf("y"));//3-->index value.
    System.out.println(s1.lastIndexOf("y"));// repeated text me last text ka index.
    System.out.println(s1.startsWith("Pr"));//true-> case sensative.
    System.out.println(s1.endsWith("sh"));//true-> case sensative.
// Extraction / Transformation.
    System.out.println(s1.substring(1, 4));// "riy"=>isme endIndex "4"excluded hota ha.
    System.out.println(s1.toUpperCase());//PRIYANSH
    System.out.println(s1.toLowerCase());//priyansh
    System.out.println(s1.trim());//it trims only spaces from start and last, only for ASCII values.
    System.out.println(s1.strip());// same as trim(); but UNICODE friendly.
    System.out.println(s1.repeat(4));//repeats Priyansh 4 times.
    System.out.println(s1.replace('i', 'p'));//Prpyansh
```

```java
        System.out.println(s1.replaceAll("ri", "uz"));// jitne bhi "ri" hoge sabhi replace ho jayge.
    // Split Method-->it splits Arrays.
        String s3 = "Ram, Raghav, Saurav, Rinku, Yash";
            String[] arr = s3.split(",");
                for(String s : arr){
                    System.out.println(s);// sabhi names alag-alag line me aayege.
                }
    //join method=> it is static method.
        System.out.println(String.join("-", "a", "b", "c"));//a-b-c
    // Conversion. from int--> String.
        String s4 = new String(String.valueOf(10));
            System.out.println(s4);//"10"
    // to get string's UNICODE values.
        byte[] arr1 = s1.getBytes();
            for(byte i : arr1){
                System.out.print(i + "-");// sabki UNICODE values aayegi.
            }
    // Advance method. intern();
        String s5 = new String("Hey Ram");
        String s6 = s5;
        String s7 = s5.intern();// ye s5 ki value ko heap memory se String pool wali memory me copy krega.
    // Advance method. format(); it is Static method.
        String name = "Priyansh";
        int age = 28;
            System.out.println("Hello "+ name + ", your age is " + age );// but this is not readable.
            System.out.println(String.format("Hello %s, your age is %s", name, age));
```

Constructor overload Methods in String

```java
String s1 = new String();// Empty string.
    String s2 = new String(""); // also same empty String.
        System.out.println(s1);         System.out.println(s2);
    String s3 = new String("Hello");//ye heap me store hoga, or iski ek copy String pool me hogi.
        System.out.println(s3);
                                    // Char Array--->
    char[] arr = {'A', 'd', 'i', 't', 'y', 'a',};
    String s6 = new String(arr); // to print all array.
        System.out.println(s6);//ans-->Aditya.
                                    //char Array Subset--->
    // offset-->Array ki place value, 1st array block is always at 0th place.
    // count-->counting, always starts from 1, 2, 3....
    String s7 = new String(arr, 0, 4);//Ans--> Adit-> from 0th array aount 4 digits.
        System.out.println(s7);
                                    // byte Array subset-->
    byte[] arr2 = {94, 95, 96, 97};
    String s8 = new String(arr2, 3, 1);
        System.out.println(s8);
                                    //StringBuilder-->
    StringBuilder sb = new StringBuilder("Hello");
    String s9 = new String(sb);
        System.out.println(s9);// result--> Hello.
                                    // StringBuffer-->
    StringBuffer sbf = new StringBuffer("bye yr");
    String s10 = new String(sbf);
        System.out.println(s10);
```

StringBuilder / StringBuffer

- **Both are mutable, it means it can change things in same string.**
- **These both are the classes, and these inherits from parent class “ AbstractStringBuilder”.**
- **These classes are also present in java.lang package, as same as String class.**
- **Both classes have same methods except,**

- **StringBuilder→ it is not thread safe.**
- **StringBuffer→ it is thread safe.**
- **StringBuilder is faster then StringBuffer.**
- **StringBuilder is used more then StringBuffer because StringBuffer is slower.**

StringBuilder/ StringBuffer :

- **Ye values ke array ko bada rkta ha.**
- **Default arry ki values capacity 16 hoti ha.**
- **Agar values capacity bhar jaye to fir vo (initial capacity * 2) + 2 formula use krta ha, to capacity ho jaygi → (16*2+2=34) ans.**
- **These, do not override the 'equals();'.**

<u>StringBuilder class</u>→ **Instance variables→**	**Class StringBuilder extends AbstractStringBuilder{** ● **Byte[ ] values;** ● **Int Count;** → ye count krta ha ki capacity me kitna space bhar gaya ha. **}**
constructor in StringBuilder/ StringBuffer→	● **new StringBuilder();** // empty constructor. ● **new StringBuilder(50);** //default array capacity 50 ho jaygi. ● **new StringBuilder("java");** // capacity=> 16 + 4= 20.
Methods in StringBuilder/ StringBuffer—>	● **append();** ● **insert();** ● **delete();** ● **replace();** ● **reverse();** ● **charAt();** ● **setcharAt();** ● **length();** ● **capacity();** ● **Ensure capacity(100);** ● **trimToSize()**;// empty spaces or capacity ko trim kr deta h jitna array ka size hota ha.